

pandas.plotting subpackage

Pandas provides some extra plotting functions for a few select plot types.

About the Data

In this notebook, we will be working with Facebook's stock price throughout 2018 (obtained using the stock_analysis package).

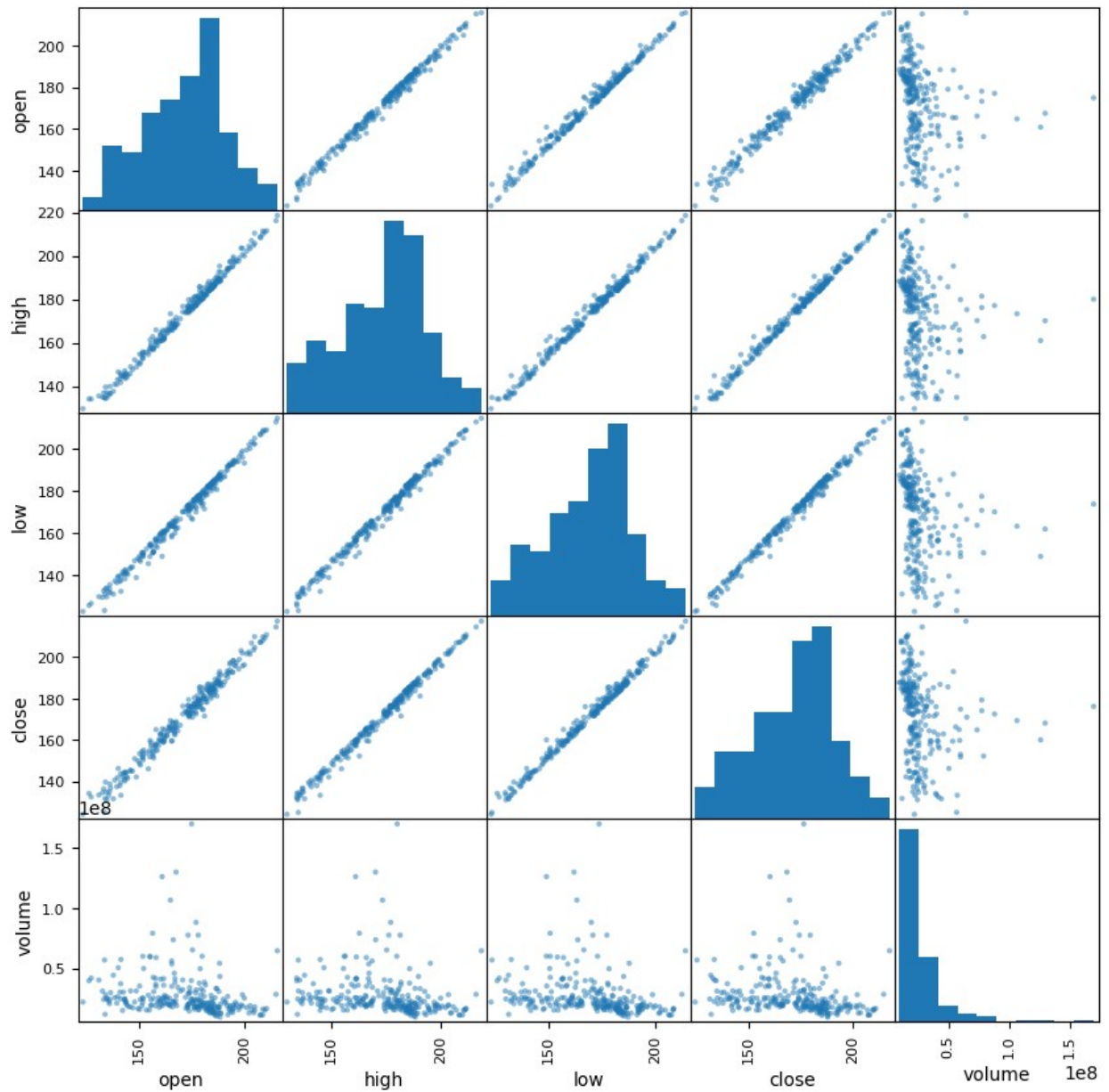
Setup

```
%matplotlib inline
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

# Load the data
fb = pd.read_csv('fb_stock_prices_2018.csv', index_col='date',
parse_dates=True)
```

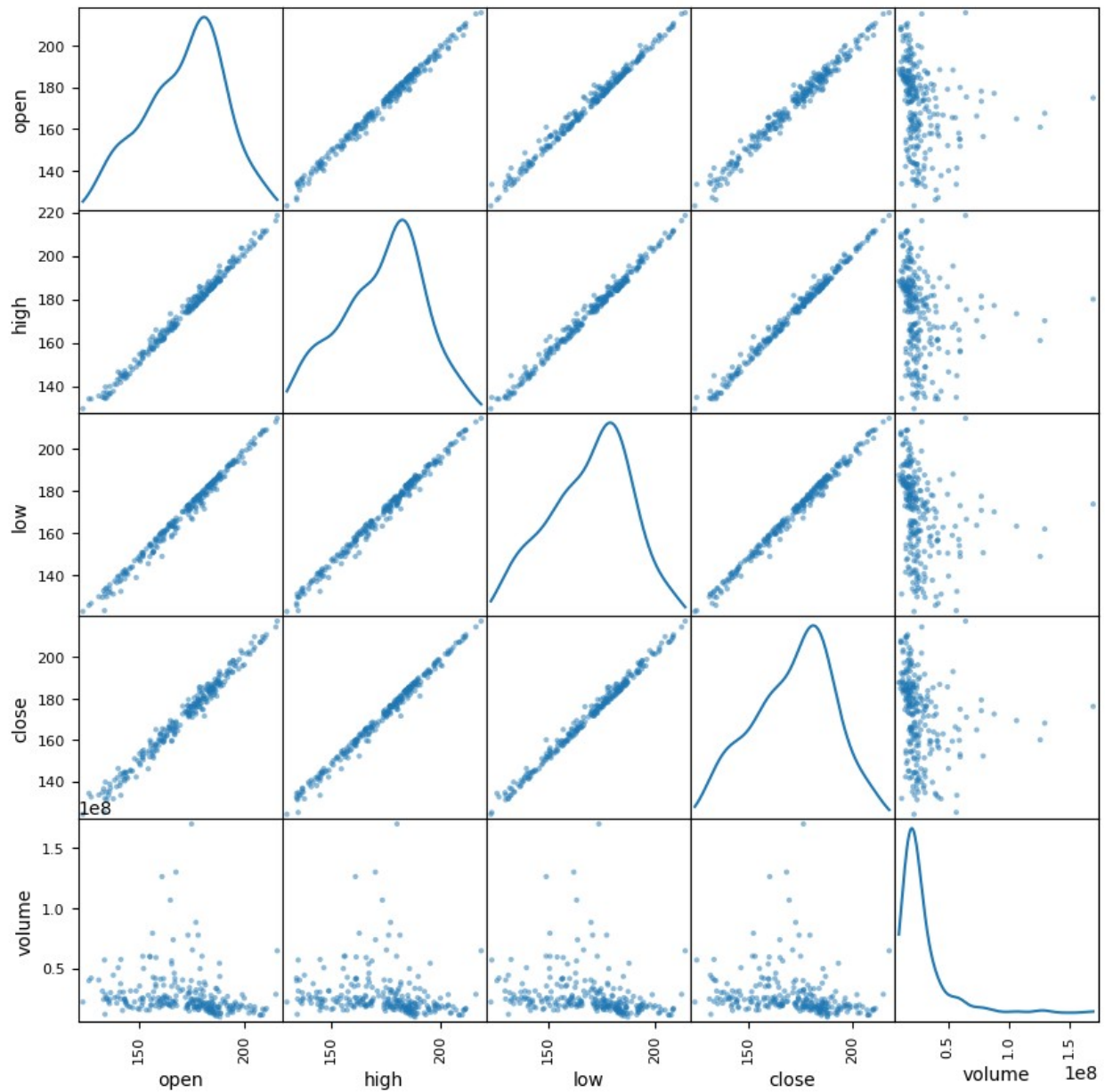
Scatter matrix

```
from pandas.plotting import scatter_matrix
# Creates a grid of scatter plots for every numeric column to see
relationships between variables
# The diagonal shows histograms of each individual column's
distribution
scatter_matrix(fb, figsize=(10, 10));
plt.show()
```



Changing the diagonal from histograms to KDE:

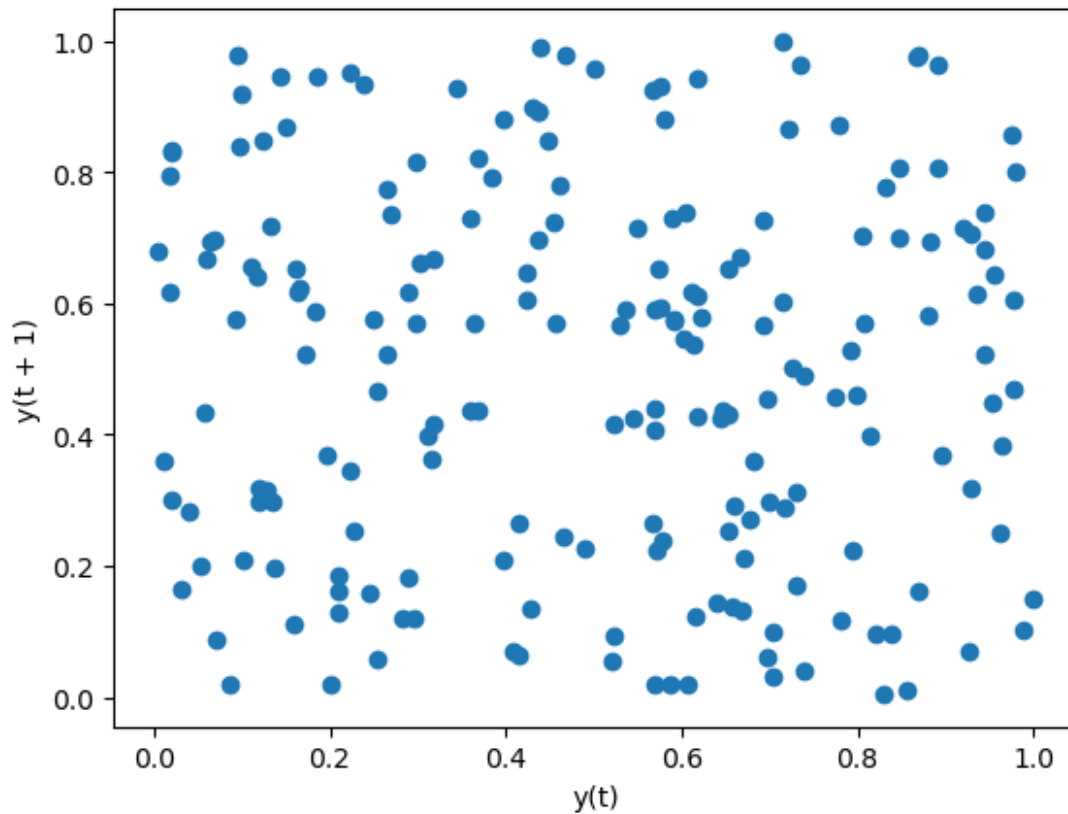
```
# Re-creates the matrix but replaces histograms with Kernel Density
# Estimation (KDE)
# to show a smoothed version of the data distribution on the diagonal
scatter_matrix(fb, figsize=(10, 10), diagonal='kde')
plt.show()
```



Lag plot

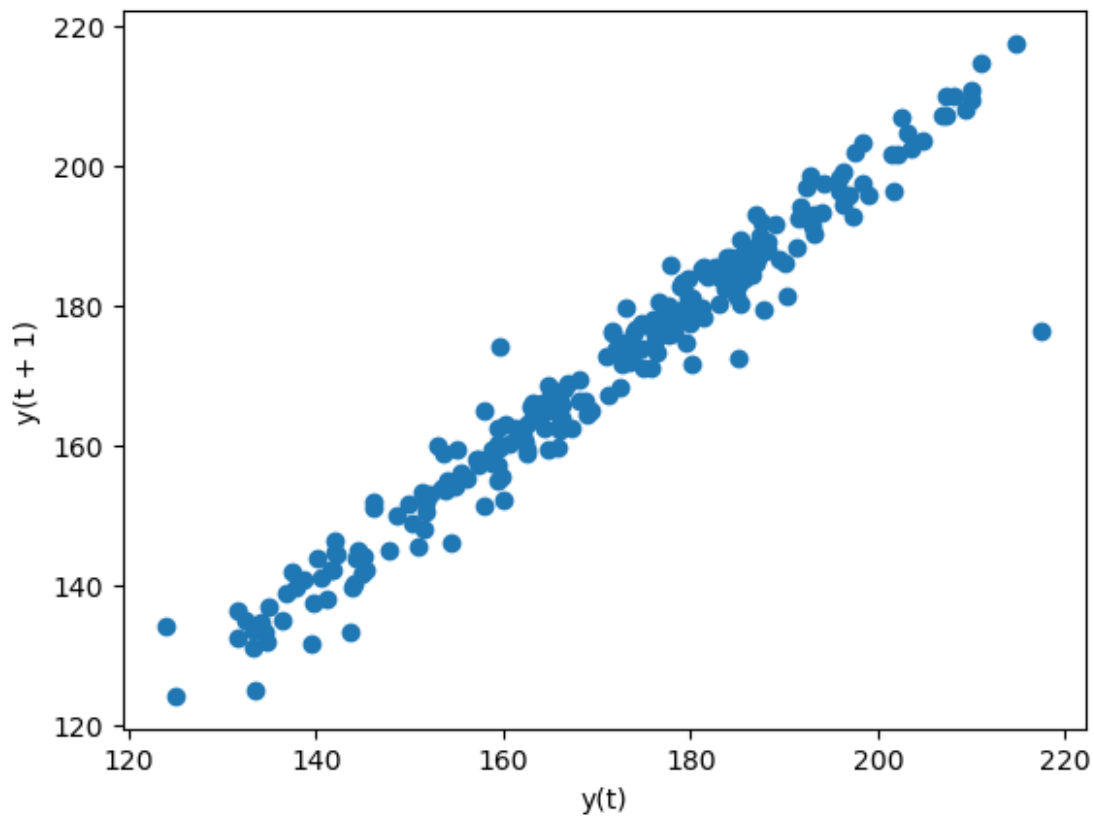
Lag plots let us see how the variable correlations with past observations of itself. Random data has no pattern:

```
from pandas.plotting import lag_plot
np.random.seed(0) # Makes the results repeatable
lag_plot(pd.Series(np.random.random(size=200)))
plt.show()
```



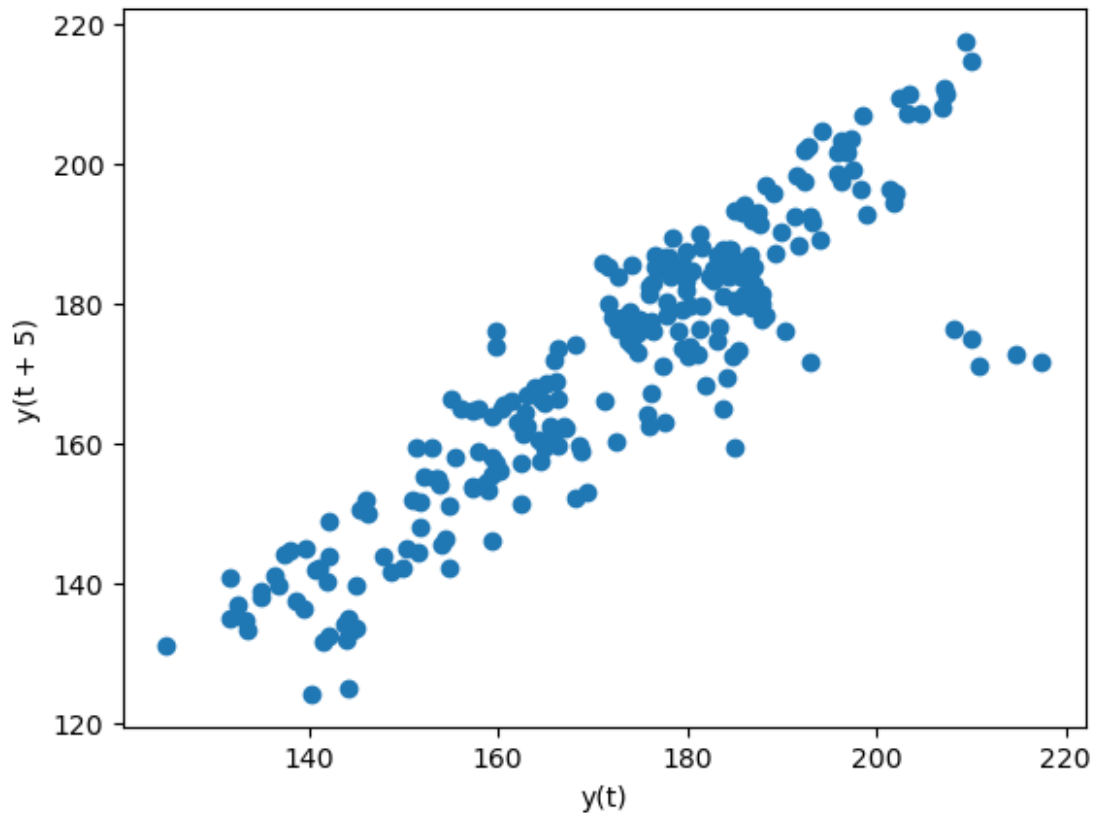
Data with some level of correlation to itself (autocorrelation) may have patterns. Stock prices are highly auto-correlated:

```
# A lag plot of stock prices usually forms a tight diagonal line  
# This indicates high autocorrelation: today's price is very similar  
to yesterday's  
lag_plot(fb.close)  
plt.show()
```



The default lag is 1, but we can alter this with the lag parameter. Let's look at a 5 day lag (a week of trading activity):

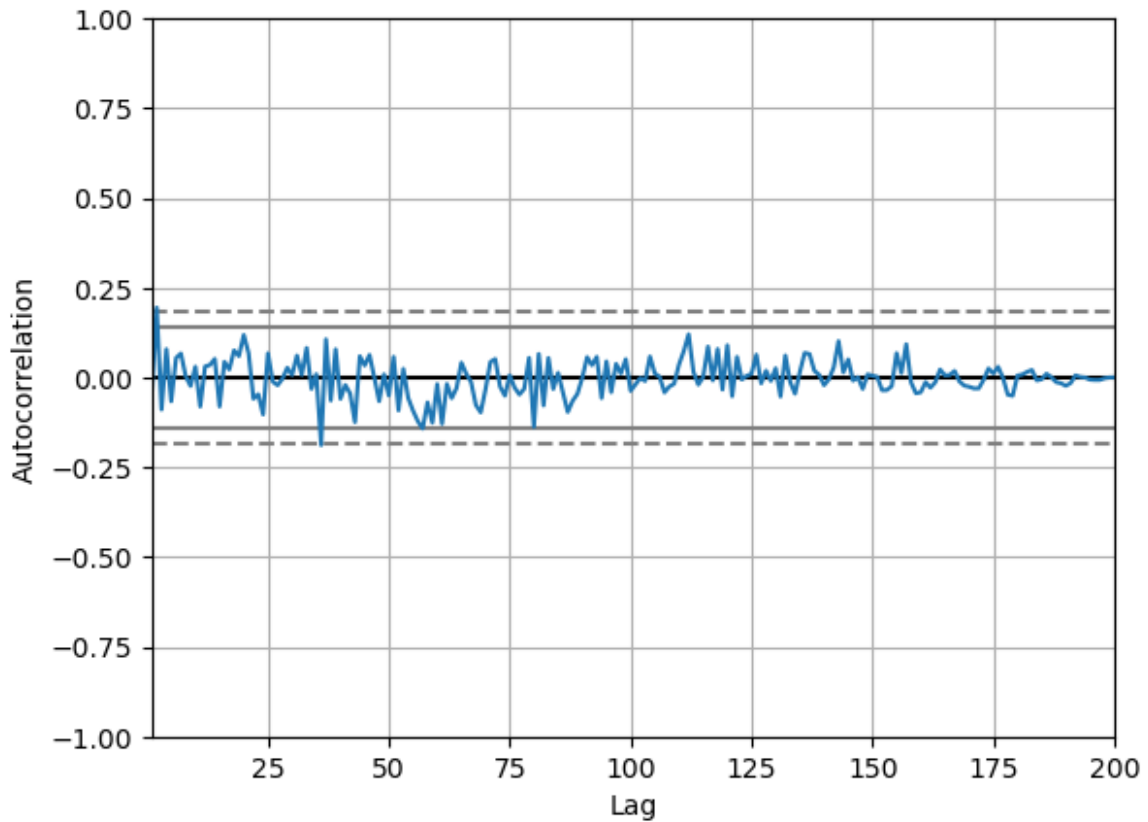
```
# Changing the lag to 5 days (one trading week)  
# Even with a longer gap, stock prices often still show a strong  
linear relationship  
lag_plot(fb.close, lag=5)  
plt.show()
```



Autocorrelation plots

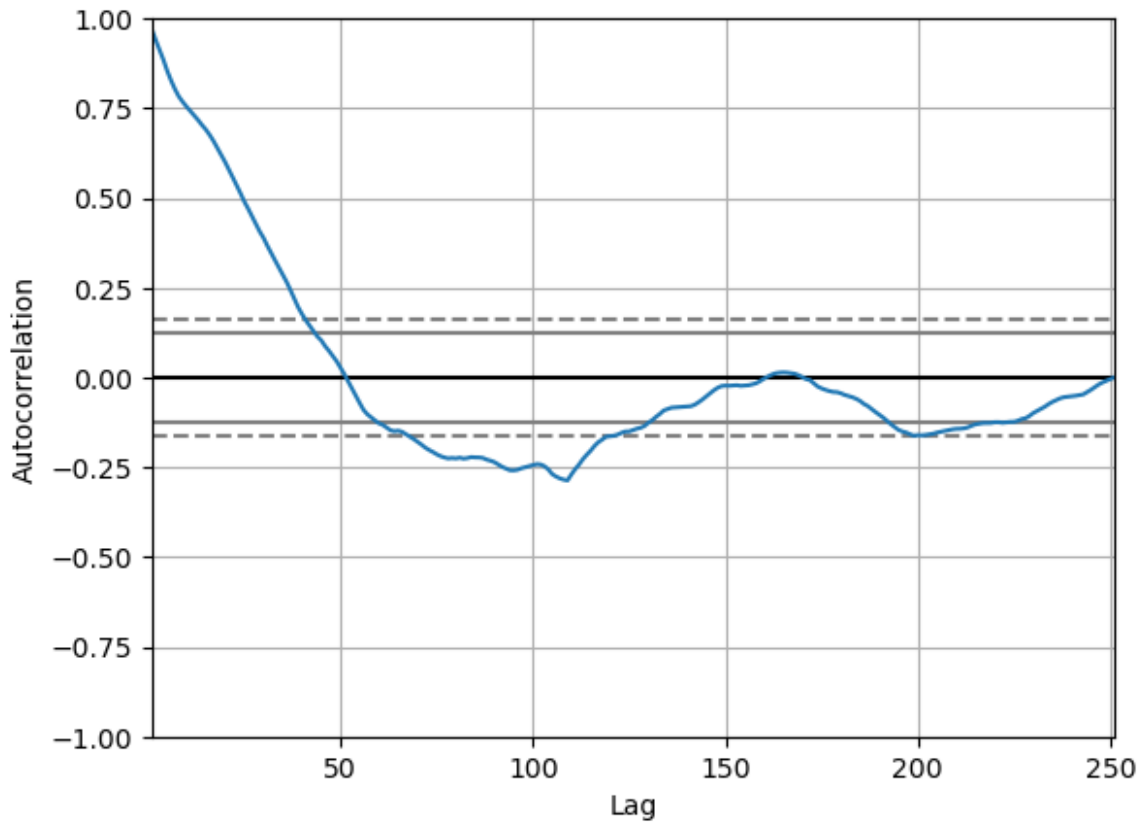
We can use the autocorrelation plot to see if this relationship may be meaningful or just noise. Random data will not have any significant autocorrelation (it stays within the bounds below)

```
from pandas.plotting import autocorrelation_plot
np.random.seed(0)
# Autocorrelation plots show how much a variable correlates with
# itself over many lags
# Random data stays near zero and within the dashed confidence bands
# (noise)
autocorrelation_plot(pd.Series(np.random.random(size=200)))
plt.show()
```



Stock data, on the other hand, does have significant autocorrelation:

```
# Stock data starts with high autocorrelation that slowly decays  
# This shows the data is 'meaningful' and not just random fluctuations  
autocorrelation_plot(fb.close)  
plt.show()
```



Bootstrap plot

This plot helps us understand the uncertainty in our summary statistics:

```
from pandas.plotting import bootstrap_plot
# Randomly resamples the data to calculate summary statistics many
times
# This helps visualize the stability/uncertainty of the mean and
median of trading volume
fig = bootstrap_plot(fb.volume, fig=plt.figure(figsize=(10, 6)))
plt.show()
```